Organic synthesis

This topic pulls the organic chemistry of the course together. It shows how to plan a route from one compound to another, how to lengthen a carbon chain with Grignard reagents, and the practical techniques used to make, separate, purify and identify an organic product safely.

Planning a synthetic route

To plan a synthesis, identify the functional group you start with and the one you need, then work out a sequence of known reactions that connects them, often working backwards from the target. For each step you must be able to give the **reagents**, the **conditions** and the **type of reaction**. Fewer steps usually means a higher overall yield, because some product is lost at every stage.

Key aliphatic conversions

From → to	Reagents and conditions	Type of reaction
alkane → halogenoalkane	Cl_2 or Br_2, UV light	free-radical substitution
alkene → alkane	H_2, nickel catalyst, heat	addition (reduction)
alkene → halogenoalkane / dihalogenoalkane	HBr or Br_2, room temperature	electrophilic addition
alkene → alcohol	steam, phosphoric(V) acid catalyst, high temperature and pressure	hydration (addition)
alkene → diol	acidified potassium manganate(VII)	oxidation
halogenoalkane → alcohol	aqueous NaOH or KOH, heat under reflux	nucleophilic substitution
halogenoalkane → alkene	KOH in ethanol, heat under reflux	elimination
halogenoalkane → nitrile	KCN in aqueous ethanol, heat under reflux	nucleophilic substitution (adds one carbon)
halogenoalkane → amine	excess ammonia in ethanol, heat in a sealed tube	nucleophilic substitution

alcohol → halogenoalkane	PCl_5; or KBr with 50% H_2SO_4; or red phosphorus and iodine	substitution
alcohol → alkene	concentrated H_3PO_4 (or H_2SO_4), heat	elimination (dehydration)
primary alcohol → aldehyde	$K_2Cr_2O_7$ and dilute H_2SO_4, heat gently and distil off the aldehyde as it forms	oxidation
primary alcohol or aldehyde → carboxylic acid	excess $K_2Cr_2O_7$ and dilute H_2SO_4, heat under reflux	oxidation
secondary alcohol → ketone	$K_2Cr_2O_7$ and dilute H_2SO_4, heat under reflux	oxidation
aldehyde, ketone or carboxylic acid → alcohol	$LiAlH_4$ in dry ether	reduction
aldehyde or ketone → hydroxynitrile	HCN with KCN	nucleophilic addition (adds one carbon)
nitrile → carboxylic acid	dilute HCl, heat under reflux	hydrolysis
nitrile → amine	$LiAlH_4$ in dry ether, or H_2 with a nickel catalyst	reduction
carboxylic acid → acyl chloride	PCl_5, room temperature	substitution
carboxylic acid → ester	alcohol with concentrated H_2SO_4, heat	esterification (condensation)
acyl chloride → acid / ester / amide / N-substituted amide	water / alcohol / concentrated ammonia / primary amine, all at room temperature	addition–elimination
ester → carboxylic acid (or its salt) + alcohol	dilute acid or aqueous NaOH, heat under reflux	hydrolysis

Aromatic routes

- **Benzene** → **nitrobenzene** → **phenylamine**: concentrated HNO_3 and H_2SO_4 below 55 °C; then tin and concentrated HCl, heat, followed by NaOH.
- **Phenylamine** → **azo dye**: $NaNO_2$ and HCl below 10 °C, then phenol in NaOH. Or phenylamine + an acyl chloride gives an N-phenyl amide.
- **Benzene** → **phenylethanone** (CH_3COCl with $AlCl_3$), which then reacts like any ketone: $LiAlH_4$ gives 1-phenylethanol, which can be esterified; HCN/KCN gives a hydroxynitrile, which hot HCl hydrolyses to a hydroxy acid.

- **Other ring reactions**: Br_2/$FeBr_3$, a chloroalkane with $AlCl_3$, SO_3 in fuming sulfuric acid, and H_2 over nickel (see Topic 18).

Grignard reagents

A **Grignard reagent**, RMgX, is made by warming a halogenoalkane with magnesium turnings in **dry ether**: $CH_3CH_2Br + Mg \rightarrow CH_3CH_2MgBr$ (ethylmagnesium bromide). The carbon bonded to magnesium carries a partial negative charge, so it acts as a **nucleophilic carbon** and forms new carbon–carbon bonds, lengthening the chain. Everything must be dry, because water destroys the reagent.

The Grignard reagent adds to the δ+ carbon of a C=O group; adding water (or dilute acid) afterwards gives the product.

Grignard reagent reacts with	Product (after adding water)	Example with CH_3CH_2MgBr
methanal, HCHO	primary alcohol	$CH_3CH_2CH_2OH$, propan-1-ol
another aldehyde, R′CHO	secondary alcohol	with ethanal: $CH_3CH_2CH(OH)CH_3$, butan-2-ol
a ketone, R′COR″	tertiary alcohol	with propanone: $CH_3CH_2C(OH)(CH_3)_2$, 2-methylbutan-2-ol
carbon dioxide	carboxylic acid	CH_3CH_2COOH, propanoic acid

The general equations, with the magnesium ending up as Mg(OH)X: RMgX + HCHO → RCH_2OH; RMgX + R′CHO → RCH(OH)R′; RMgX + R′COR″ → RR′C(OH)R″; RMgX + CO_2 → RCOOH (each after adding H_2O). Other ways to add a carbon are reacting a halogenoalkane with KCN, or a carbonyl with HCN.

Practical techniques

Heating under reflux and distillation

- **Reflux** heats a reaction mixture for a long time without losing volatile substances: a vertical condenser turns vapours back to liquid, which drips back into the flask. It is used, for example, to oxidise a primary alcohol fully to a carboxylic acid with excess dichromate. **Never seal the top** of the condenser: pressure would build up and the apparatus could explode.
- **Distillation** separates a liquid product from the mixture, collecting the fraction that boils near the product's boiling point; the thermometer bulb sits level with the side arm. Distilling the aldehyde off as it forms, with limited dichromate, stops it being oxidised further.
- **Anti-bumping granules** make small bubbles form, preventing violent uneven boiling. Organic liquids are heated with electric heaters or water baths, as they are usually flammable.

- When drawing apparatus: no lines across the joins between flask, still head and condenser; the condenser has an outer water jacket sealed at both ends, with water in at the bottom and out at the top.

Purifying an organic liquid

1. Put the impure product in a **separating funnel** and shake it with sodium hydrogencarbonate solution to neutralise acid impurities, releasing the pressure of CO_2 often. Shaking with saturated sodium chloride solution helps the layers separate.
2. Let the layers separate and run off the aqueous layer. The organic layer is usually the less dense, upper layer, but check densities.
3. Run the organic layer into a dry flask and add a **drying agent**, such as anhydrous calcium chloride or magnesium sulfate, until the liquid is clear. The drying agent must be insoluble in, and not react with, the product.
4. Decant or filter the liquid off the drying agent and **distil**, collecting the fraction at the product's boiling point.

Purifying an organic solid: recrystallisation

Step	Reason
Dissolve the impure solid in the **minimum volume of hot solvent**.	The solvent should dissolve the compound well when hot but poorly when cold. Using the minimum gives a saturated solution, so crystals form on cooling.
Filter the hot solution quickly through a fluted filter paper.	Removes insoluble impurities; keeping it hot stops crystals forming in the funnel.
Cool the filtrate, finally in ice.	The compound crystallises; soluble impurities stay dissolved because there is too little of them to saturate the solution.
Collect the crystals by suction filtration with a Büchner funnel.	Reduced pressure makes filtration fast.
Wash with a little cold solvent and dry between filter papers.	Removes soluble impurities on the surface without dissolving much product. Wet crystals give a mass that is too high, even a "yield" above 100%.

Steam distillation and solvent extraction

- **Steam distillation**: steam is blown through the mixture and the product distils over with the water at below 100 °C, so a high-boiling compound that would decompose at its own boiling point can be separated. Phenylamine is purified this way.
- **Solvent extraction**: shake the mixture with an organic solvent in which the product dissolves better than in water, separate the layers, dry the organic layer and distil off the solvent.

Checking purity

- **Melting point**: a pure solid melts sharply at the value in data books; impurities lower the melting point and make it melt over a range. Use a melting-point apparatus, or a capillary tube strapped to a thermometer in an oil bath (the oil must boil above the sample's melting point). Heat slowly so the thermometer and sample stay at the same temperature.
- **Boiling point**: measured with the thermometer bulb in the vapour just above the boiling liquid. It is less useful for identification, since different liquids can share a boiling point, and it changes with atmospheric pressure.

Hazards and risks

- A **hazard** is anything with the potential to cause harm: toxic, flammable, harmful, irritant, corrosive, oxidising, carcinogenic. A **risk** is the chance that harm actually results, which depends on the amount, the concentration and how the substance is used.
- Control measures match the hazard: eye protection for irritants and corrosives, gloves and a fume cupboard for toxic substances, no naked flames near flammable liquids, oxidising agents kept away from flammable materials.

Identifying the product

Combustion analysis

Empirical formula from combustion

1. 0.290 g of a compound of C, H and O burns completely to give 0.660 g of CO_2 and 0.270 g of H_2O. Find its empirical formula.
2. Carbon: moles of CO_2 = 0.660 ÷ 44.0 = 0.0150 mol, so 0.0150 mol of C, mass 0.180 g.
3. Hydrogen: moles of H_2O = 0.270 ÷ 18.0 = 0.0150 mol, so 2 × 0.0150 = 0.0300 mol of H, mass 0.030 g.
4. Oxygen by difference: 0.290 − 0.180 − 0.030 = 0.080 g, so 0.080 ÷ 16.0 = 0.0050 mol.
5. Ratio C : H : O = 0.0150 : 0.0300 : 0.0050 = 3 : 6 : 1, so the empirical formula is $\mathbf{C_3H_6O}$. If the mass spectrum shows M^+ at 58, this is also the molecular formula: propanone or propanal, which IR, NMR or chemical tests can tell apart.

The spectroscopic methods (mass spectra, IR and NMR) are covered in Topics 10 and 15, including a full worked identification.

Chemical tests for functional groups

Functional group	Test	Positive result
alkene	bromine water	orange to colourless
alcohol or carboxylic acid (–OH)	PCl_5	steamy fumes of HCl
alcohol, phenol or carboxylic acid	sodium metal	fizzing (hydrogen)
aldehyde or ketone	2,4-DNPH	orange precipitate
aldehyde	Fehling's or Benedict's solution, warm	blue to brick-red precipitate
aldehyde	Tollens' reagent, warm	silver mirror
$CH_3CO–$ or $CH_3CH(OH)–$	iodine and NaOH, warm	pale yellow precipitate of CHI_3
carboxylic acid	sodium carbonate or hydrogencarbonate	fizzing (CO_2)
primary or secondary alcohol, aldehyde	acidified $K_2Cr_2O_7$, warm	orange to green
halogenoalkane	warm with NaOH(aq), acidify with HNO_3, add $AgNO_3$	white (Cl), cream (Br) or yellow (I) precipitate, forming slowly
acyl chloride	water, then $AgNO_3$	steamy fumes, then an immediate white precipitate

Quick checks

QUICK CHECK

Which sequence converts bromoethane into propanoic acid?

- aqueous NaOH, then acidified dichromate
- KCN in ethanol, then heat with dilute HCl
- ammonia, then $LiAlH_4$
- KOH in ethanol, then steam

Cyanide adds a carbon (CH_3CH_2CN), and acid hydrolysis turns the nitrile into CH_3CH_2COOH. NaOH then oxidation would give ethanoic acid, with only two carbons.

QUICK CHECK

Which product forms when CH_3MgI reacts with propanone and the mixture is then treated with water?

- propan-2-ol
- butan-2-ol
- 2-methylpropan-2-ol
- butanoic acid

A Grignard reagent with a ketone gives a tertiary alcohol: the CH_3 joins the carbonyl carbon, giving $(CH_3)_3COH$.

QUICK CHECK

Why is a primary alcohol oxidised under reflux to make a carboxylic acid, but distilled to make an aldehyde?

- reflux removes water
- reflux keeps the aldehyde in the flask to be oxidised further; distilling removes it as soon as it forms
- distillation is faster
- the aldehyde needs a higher temperature

The aldehyde has a lower boiling point than the alcohol and acid, so distillation carries it away before it meets more oxidising agent.

QUICK CHECK

In recrystallisation, why is the minimum volume of hot solvent used?

- to save solvent
- to dissolve the impurities
- so the solution is saturated and the product crystallises on cooling
- so the solution cools faster

With too much solvent, much of the product stays dissolved when cold and the yield falls, or no crystals form at all.

QUICK CHECK

A student's crystals melt between 118 °C and 124 °C; the data book value is 127 °C. What does this suggest?

- the crystals are pure
- the crystals are impure: impurities lower the melting point and widen the range
- the thermometer reads too high
- the compound has decomposed into a purer form

A pure substance melts sharply at its data book value. A low, broad melting range points to impurities, such as remaining solvent.

QUICK CHECK

Which statement about hazard and risk is correct?

- a hazard is the chance of harm
- a dilute solution of a corrosive substance has the same risk as the concentrated one
- risk depends on how much of a hazardous substance is used and how
- flammable liquids are a risk only when heated with electric heaters

A hazard is the potential to cause harm; risk is the chance that harm actually happens, which falls with smaller amounts, lower concentrations and good control measures.